

Exact Event Maps and Cluster Coarsening in Pulse-Coupled Integrate-and-Fire Networks

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Abstract

We give a finite exact event certificate for an all-to-all excitatory network of identical integrate-and-fire oscillators. The strictly concave rise $U_a(\phi) = (1 - e^{-a\phi})/(1 - e^{-a})$ becomes rational event arithmetic in $u = e^{-a\phi}$ when $r = e^{-a}$ is rational. Free flight is a common rescaling, a pulse subtracts $(1 - r)\epsilon$ with threshold clipping, and same-time avalanche closure is explicit. We prove equality-partition coarsening and the absorbing synchronized primitive event cycle $[N]$. The receipt contains 441 finite probes for $N \leq 8$, an independent checker, symbolic checks, replay, and hostile mutations. Mirollo–Strogatz synchrony is cited separately; no exhaustive continuous-state census or target spectral claim is made. The strict Route-A verdict is `ROUTE_A_REJECTED`.

1 Model and event coordinates

Let $N \geq 2$ phases satisfy $\dot{\phi}_i = 1$ on $[0, 1)$. With $r = e^{-a} \in (0, 1)$, define

$$U_a(\phi) = \frac{1 - e^{-a\phi}}{1 - r}, \quad u_i = e^{-a\phi_i}. \quad (1)$$

Threshold $U_a = 1$ is $u = r$, and reset is $U_a = 0$, $u = 1$. A firing oscillator sends an excitatory increment $\epsilon \in (0, 1)$ to every nonfiring oscillator in the U -coordinate. Pulses are closed at the same time: a newly thresholded oscillator fires and sends one further pulse.

Theorem 1 (Exact event map). *For a post-avalanche state $u \in [r, 1]^N$, free flight to the next event is*

$$\hat{u}_i = \frac{r}{\min_j u_j} u_i. \quad (2)$$

The threshold set is $F_0 = \{i : \hat{u}_i = r\}$. If F_k is one simultaneous recruitment wave, every nonfiring coordinate is updated by

$$u_j \leftarrow u_j - (1 - r)\epsilon |F_k|, \quad F_{k+1} = \{j \notin F_0 \cup \dots \cup F_k : u_j \leq r\}, \quad (3)$$

then all indices in the finite closure $F = \bigcup_k F_k$ reset to one.

Proof. During free flow $u_i(t) = u_i(0)e^{-at}$; solving $e^{-at} \min_i u_i = r$ gives (2). Since $U_a^{-1}(y) = \log(1 - (1 - r)y)/\log r$, adding ϵ in y subtracts $(1 - r)\epsilon$ in u , with clipping at r . Applying all pulses in a wave simultaneously yields (3). Each wave adds at least one new index, so the closure terminates in at most N waves. $\square$

Proposition 1 (Rationality and no splitting). *If r, ϵ , and the event state are rational, every event state is rational. Moreover, if two labelled coordinates are equal before an event, they are equal after it; equivalently every old equality block is contained in a unique new equality block.*

Proof. All operations in (2)–(3) are rational comparisons and additions/multiplications. Common scaling preserves equality. A nonfiring equality block receives the same subtraction in every wave; if it reaches threshold, the whole block is recruited and reset together. Thus no old block can split, even though different blocks may merge during an avalanche. $\square$

2 Synchronized cycle and finite certificate

At $u = (1, \dots, 1)$, (2) gives $u = (r, \dots, r)$; all N indices fire, and reset returns to $(1, \dots, 1)$. The event word is the one-letter word $[N]$, hence primitive of period one. This is a source-local event cycle, not an arithmetic primitive orbit.

The exact ledger uses $r \in \{1/2, 2/3, 3/4\}$, $\epsilon \in \{1/5, 1/4, 1/3\}$, $N = 2, \dots, 8$, seven deterministic rational seeds, and twelve event steps: 441 rows. It retains 63 all-equal synchronized rows and records every recruitment wave, old-block containment receipt, and cycle label. It intentionally does not enumerate all continuous parameter cells or all initial states.

Table 1: Representative exact event rows. The event word lists firing cardinalities; recruitment waves are serialized separately in the receipt.

r	ϵ	N	seed	event word	cycle
1/2	1/5	2	0	[2]	1
2/3	1/4	4	0	[4]	1
3/4	1/3	8	0	[8]	1
1/2	1/5	2	1	[1,1,1,1,1,1,1,1,1,1]	–

3 Literature boundary and audit receipt

Mirollo and Strogatz establish almost-everywhere synchrony for identical integrate-and-fire oscillators with a strictly concave rise and all-to-all excitatory pulses [1]. Bottani studies globally coupled synchronization beyond concavity/convexity restrictions [2]. We preserve the hypotheses and the almost-everywhere qualifier; the finite receipt is independent evidence, not a reproof or a complete continuous-state cell census. Simultaneous ties, $\epsilon = 0$, $r \rightarrow 1^-$, $r \rightarrow 0^+$, directed coupling, and inhibitory pulses are explicit boundaries.

The independent checker passes 4,438 assertions, SymPy passes 330 identities, byte replay matches two fresh producer runs, and 41 repaired/stale-hash mutations are rejected. Every old equality block is checked by explicit containment, not merely by a block-count comparison.

References

- [1] R. E. Mirollo and S. H. Strogatz, “Synchronization of Pulse-Coupled Biological Oscillators,” *SIAM Journal on Applied Mathematics* 50 (1990), 1645–1662, DOI: 10.1137/0150098.
- [2] S. Bottani, “Synchronization of integrate and fire oscillators with global coupling,” *Physical Review E* 54 (1996), 2334–2340, DOI: 10.1103/PhysRevE.54.2334.

Declarations

Scope. NO_BAD_EULER_OR_ROOT_NUMBER; no arithmetic, target-zero, Euler-factor, root-data, automorphy, or Hilbert–Pólya claim is made. **Data and code.** The exact receipt, independent checker, replay, and mutation audit are included. **AI-use disclosure.** Generative tools assisted drafting and code generation; formulas and metadata are checked by the deterministic artifact chain.